

**Source
Initialization**

**BFS
Traversal**

**Distance and
Path Counting**

**Reverse
Dependency
Accumulation**

**BC
Accumulation**

**Next
Source**

Repeat for every source vertex

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graph LR; A[Source Initialization] --> B[BFS Traversal]; B --> C[Distance and Path Counting]; C --> D[Reverse Dependency Accumulation]; D --> E[BC Accumulation]; E --> F[Next Source]; F -.-> A; F -.-> G[Repeat for every source vertex];
```

The diagram illustrates a sequential process for graph algorithms. It consists of six rounded rectangular boxes arranged horizontally. The first box, 'Source Initialization', is light gray with a black border. It is connected by a blue arrow to the second box, 'BFS Traversal', which is light blue with a blue border. This is followed by a blue arrow to the third box, 'Distance and Path Counting', which is light green with a green border. A blue arrow then leads to the fourth box, 'Reverse Dependency Accumulation', which is light purple with a purple border. Another blue arrow connects to the fifth box, 'BC Accumulation', which is light orange with an orange border. Finally, a blue arrow points to the sixth box, 'Next Source', which is light gray with a black border. A dashed black line originates from the bottom of the 'Next Source' box, extends downwards, then horizontally to the left, and finally upwards with an arrowhead pointing to the bottom of the 'Source Initialization' box, forming a loop. Below the 'Next Source' box, the text 'Repeat for every source vertex' is written in bold black font.